

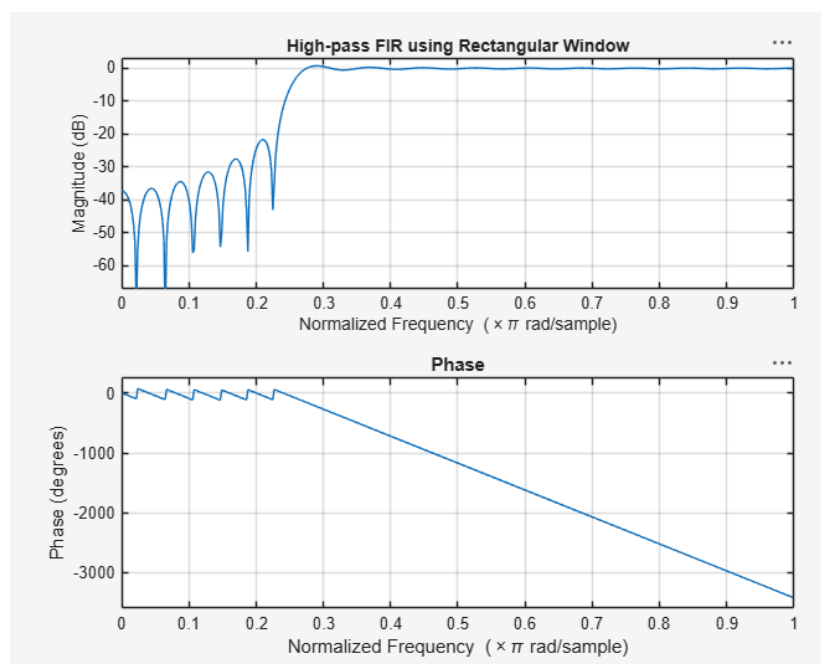
Expt 5 : Design of High Pass Filter using Windowing Techniques

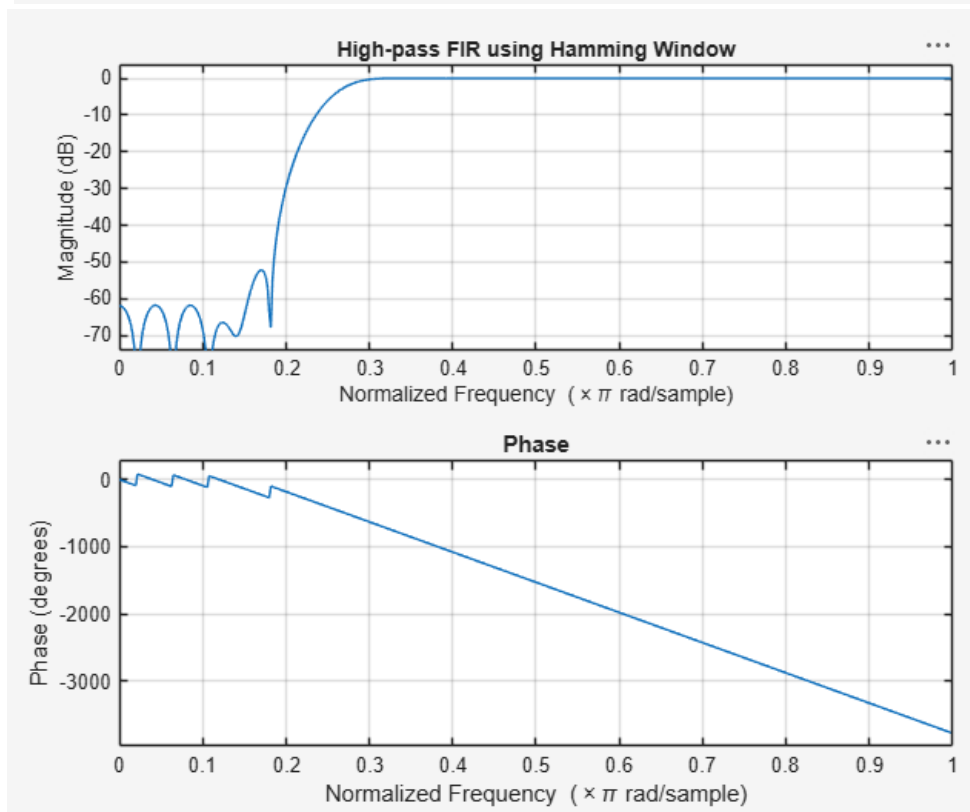
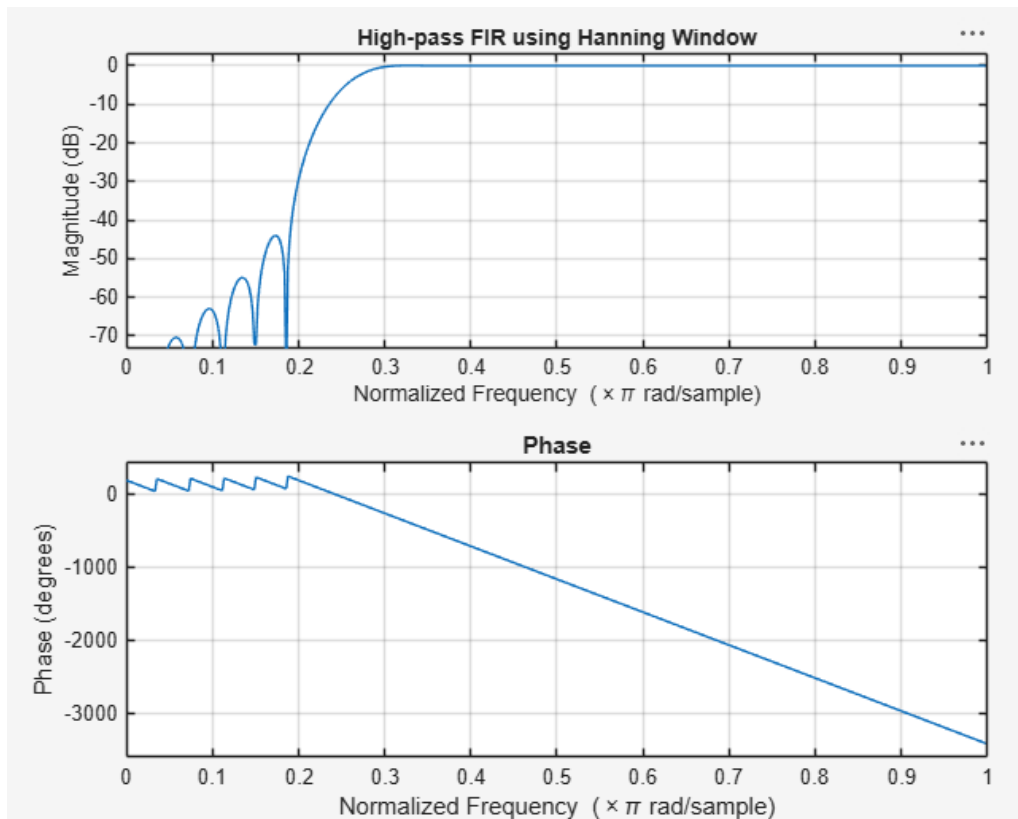
5.1 Generate the coefficients of an FIR High-Pass filter using Rectangular windowing techniques.

Program :

```
/MAILAB Drive/Usp_5a.m
1  clc; clear; close all;
2  %% --- Input Parameters ---
3  fp = 1000; % High-pass cutoff frequency (Hz)
4  fs = 8000; % Sampling frequency (Hz)
5  N = 50; % Filter order
6
7  wc = fp/(fs/2); % Normalized cutoff (0-1)
8
9  %% --- 1. FIR High Pass using Rectangular Window ---
10 b_rect = fir1(N, wc, 'high', rectwin(N+1));
11 disp('--- High-Pass Coefficients (Rectangular Window) ---');
12 disp(b_rect');
13
14 %% --- 2. FIR High Pass using Hamming Window ---
15 b_hamm = fir1(N, wc, 'high', hamming(N+1));
16 disp('--- High-Pass Coefficients (Hamming Window) ---');
17 disp(b_hamm');
18
19 %% --- 3. FIR High Pass using Hanning Window ---
20 b_hann = fir1(N, wc, 'high', hanning(N+1));
21 disp('--- High-Pass Coefficients (Hanning Window) ---');
22 disp(b_hann');
23
24 %% --- Frequency Response Plots ---
25 figure;
26 freqz(b_rect,1);
27 title('High-pass FIR using Rectangular Window');
28 figure;
29 freqz(b_hamm,1);
30 title('High-pass FIR using Hamming Window');
31 figure;
32 freqz(b_hann,1);
33 title('High-pass FIR using Hanning Window');
```

Output :





5.2. Generate the coefficients of an FIR High-Pass filter using Hamming windowing

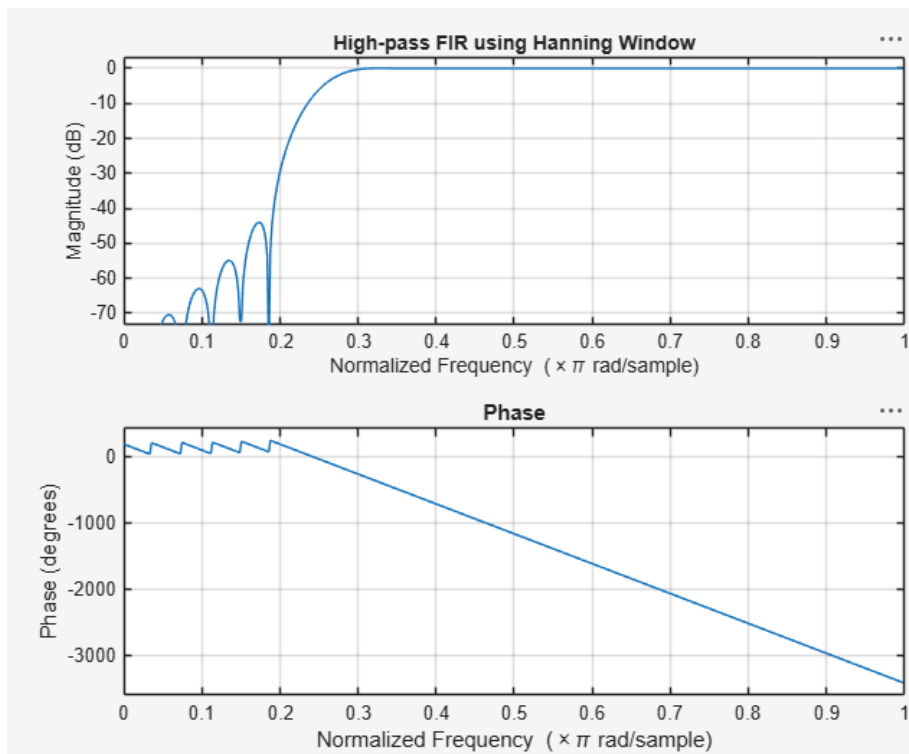
Techniques

Program:

/MATLAB Drive/Dsp_5b.m

```
1  clc; clear; close all;
2
3  %% --- Input Parameters ---
4  fp = 1000; % High-pass cutoff frequency (Hz)
5  fs = 8000; % Sampling frequency (Hz)
6  N = 50; % Filter order
7
8  wc = fp/(fs/2); % Normalized cutoff (0-1)
9  %% --- FIR High Pass using Hanning Window ---
10
11  b_hann = fir1(N, wc, 'high', hanning(N+1));
12  disp('--- High-Pass Coefficients (Hanning Window) ---');
13  disp(b_hann');
14  %% --- Frequency Response Plots ---
15  figure;
16  freqz(b_hann,1);
17  title('High-pass FIR using Hanning Window');
```

Output:



5.3. Generate the coefficients of an FIR High-Pass filter using Hanning windowing

Techniques

Program:

/MAILAB Drive/Usp_5c.m

```
1  clc; clear; close all;
2
3  %% --- Input Parameters ---
4  fp = 1000; % High-pass cutoff frequency (Hz)
5  fs = 8000; % Sampling frequency (Hz)
6  N = 50; % Filter order
7
8  wc = fp/(fs/2); % Normalized cutoff (0-1)
9  %% --- FIR High Pass using Hanning Window ---
10
11  b_hann = fir1(N, wc, 'high', hanning(N+1));
12  disp('--- High-Pass Coefficients (Hanning Window) ---');
13  disp(b_hann');
14  %% --- Frequency Response Plots ---
15  figure;
16  freqz(b_hann,1);
17  title('High-pass FIR using Hanning Window');
```

Output:

